MultiCore TECHNOLOGIES

LEARN TODAY BUILD TOMORROW

Your Future in Tech

DATA SCIENCE COURSE

Data-Driven Skills for a Smarter Tomorrow



COURSE HIGHLIGHTS

- Python, Data Science & Statistics
- SQL & Data Analytics
- Machine Learning & Predictive Modeling
- Power BI & Generative AI
- Git, Deployment & Industry Projects

DATA SCIENCE WITH GEN AI

Learn • Analyze • Build • Innovate

COURSE OVERVIEW

A practical, career-focused program covering Python programming, SQL, statistics, data analysis, machine learning, business intelligence, Generative AI, version control, deployment, and real-world projects.

WHAT WILL YOU LEARN?

PYTHON & DATA SCIENCE

- Python Programming
- NumPy & Pandas
- Data Manipulation
- Data Visualization
- Matplotlib & Seaborn

SQL & DATABASE ANALYSIS

- SQL Fundamentals
- Joins & Subqueries
- Aggregations
- CTEs & Window Functions
- Views

STATISTICS & DATA ANALYSIS

- Descriptive Statistics
- Probability
- Distributions
- Correlation & Covariance
- Hypothesis Testing

MACHINE LEARNING

- Regression & Classification
- Decision Trees & Random Forest
- KNN & Naive Bayes
- Clustering & PCA
- Model Evaluation & Tuning

GENERATIVE AI

- AI, ML & GenAI Fundamentals
- Large Language Models
- Prompt Engineering
- LLM Application Development
- RAG & Vector Databases



WHO CAN JOIN?

- Students & Graduates
- Beginners in Programming
- Aspiring Data Scientists
- Aspiring AI/ML Professionals
- Career Changers
- Working Professionals
- Data & Business Analysts
- Freelancers & Startup Enthusiasts

LEARNING OUTCOMES

By the end of the course, learners can:

- Program using Python for data applications
- Extract and analyze data using SQL
- Perform data cleaning and exploratory analysis
- Build and evaluate machine learning models
- Create interactive Power BI dashboards
- Develop AI applications using LLMs
- Build RAG-based AI chatbots
- Use Git & GitHub for project management
- Deploy data science and AI applications
- Build an industry-ready project portfolio

BUILD YOUR DATA SKILLS FOR A SMARTER TOMORROW

DATA SCIENCE WITH GEN AI

Learn. Build. Analyze. Create with AI.



PYTHON PROGRAMMING

Python Fundamentals

- Variables & Data Types
- Operators
- Conditions & Loops
- Functions & Modules
- File Handling
- Exception Handling
- Python Libraries

NumPy

- Arrays
- Mathematical Operations
- Numerical Computing

Pandas

- DataFrames
- Data Manipulation
- Data Filtering
- GroupBy & Aggregation

Visualization

- Matplotlib
- Seaborn
- Statistical Visualization

SQL & DATA ANALYSIS

SQL Fundamentals

- SELECT & WHERE
- ORDER BY
- DISTINCT & LIMIT
- Aggregate Functions

Intermediate SQL

- GROUP BY & HAVING
- INNER & LEFT JOIN
- RIGHT JOIN
- Subqueries
- CASE WHEN

Advanced SQL

- Common Table Expressions
- Window Functions
- ROW_NUMBER & RANK
- Views

KEY SKILLS

Query → Transform → Aggregate → Analyze

STATISTICS

Descriptive Statistics

- Mean • Median • Mode
- Variance
- Standard Deviation

Probability

- Probability Fundamentals
- Probability Distributions

Statistical Concepts

- Correlation
- Covariance
- Hypothesis Testing
- Feature Scaling

KEY SKILLS

Analyze → Model → Predict → Solve

BUILD REAL WORLD SKILLS

WORK ON INDUSTRY PROJECTS

EXPERT MENTORSHIP

BOOST YOUR CAREER

TURN IDEAS INTO IMPACT

DATA SCIENCE, MACHINE LEARNING & POWER BI

DATA SCIENCE WITH PYTHON

DATA CLEANING
- Missing Values
- Duplicate Handling
- Outlier Detection

DATA PREPROCESSING
- Encoding
- Scaling & Normalization
- Feature Engineering

EXPLORATORY DATA ANALYSIS
- Descriptive Statistics
- Correlation Analysis
- Data Visualization

DATA MANIPULATION
- Merge & Join DataFrames
- GroupBy & Aggregation
- Data Transformation

KEY SKILLS
Clean → Transform → Explore → Visualize → Extract Insights

MACHINE LEARNING

SUPERVISED LEARNING
- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- KNN
- Naive Bayes
- Support Vector Machine — Basics

UNSUPERVISED LEARNING
- K-Means Clustering
- Principal Component Analysis
- Dimensionality Reduction

MODEL DEVELOPMENT
- Train-Test Split
- Cross Validation
- Evaluation Metrics
- Hyperparameter Tuning — Basics

KEY SKILLS
Prepare → Train → Evaluate → Optimize → Predict

POWER BI

BUSINESS INTELLIGENCE & DASHBOARDS
- Interactive Reports
- Dashboards
- Data Visualization
- Business Insights

DATA PREPARATION
- Power Query
- Data Transformation

DATA MODELING
- Relationships
- Data Models
- DAX
- DAX Fundamentals
- Calculations & Measures

REPORTING
- Interactive Reports
- Dashboards
- Data Visualization
- Business Insights

BUILD MODERN WEB APPLICATIONS

DEVELOP SCALABLE SOLUTIONS

GAIN PRACTICAL SKILLS

BECOME INDUSTRY-READY

SKILLS TODAY
FOR A BRIGHTER TOMORROW

GENERATIVE AI, GIT, DEPLOYMENT & PROJECTS

GENERATIVE AI

AI FUNDAMENTALS
- AI, ML & Generative AI
- Large Language Models

LLM FUNDAMENTALS
- Prompt Engineering

LLM APPLICATIONS
- OpenAI API / Equivalent APIs
- API Integration
- AI-Powered Applications

LANGCHAIN
- LangChain Basics
- LLM Workflows
- Application Integration

RAG
- Retrieval-Augmented Generation
- Document-Based Question Answering
- Context Retrieval

VECTOR DATABASES
- FAISS
- ChromaDB
- Embeddings & Similarity Search

KEY SKILLS
Prompt → Retrieve → Generate → Integrate

GIT & GITHUB

VERSION CONTROL
- Git Fundamentals
- Repository Management
- Commit & Push
- Branching & Merging
- GitHub Collaboration

DEPLOYMENT
- **DEPLOY DATA & AI APPLICATIONS**

STREAMLIT
- Interactive Data Applications
- ML Model Deployment
- AI Application Deployment

FLASK
- Flask Basics
- API Development
- Application Integration

API INTEGRATION
- API Fundamentals
- Request & Response
- Connecting AI & ML Applications

KEY SKILLS
Develop → Version → Integrate → Deploy

PROJECTS
BUILD AN INDUSTRY-READY PORTFOLIO

DATA ANALYSIS PROJECT
- Data Cleaning
- Exploratory Data Analysis
- Visualization
- Business Insights

MACHINE LEARNING PROJECT
- Data Preprocessing
- Model Building
- Evaluation
- Prediction

POWER BI DASHBOARD
- Data Modeling
- DAX
- Interactive Dashboard
- Business Reporting

AI CHATBOT — RAG
- LLM Integration
- Document Retrieval
- Vector Database
- RAG Pipeline
- AI Chat Interface

FINAL OUTCOME

Analyze → Predict → Visualize → Build → Deploy

YOUR DATA SCIENCE & AI JOURNEY STARTS HERE.



- PRACTICAL LEARNING
- REAL-TIME SCENARIOS
- HANDS-ON PROJECTS
- CAREER GROWTH

SKILLS TODAY FOR A SMARTER TOMORROW